

Santosh Bala N

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EDUCATION

Vellore Institute of Technology

B.Tech Computer Science and Engineering - CGPA: 8.7

Vellore, TN

2024 – 2028

Vidya Mandir Senior Secondary School

High School Diploma - 93.4%

Chennai, TN

2010 – 2024

EXPERIENCE

Google Summer of Code 2026

Remote

Open Source Contributor - PSLab Project under FOSSASIA

May 2026 – Present

- Writing firmware in **C** with a 3-layer system architecture (platform / system / application) for a new open-source lab instrument, enabling high-throughput data capture with clean separation of hardware abstraction and application logic.
- Implementing a **logic analyzer** at **200 MSPS** (million samples/second) and oscilloscope at **80 MSPS** with industry standard instrument control protocol **SCPI** command interface over USB and **wireless UDP based streaming** for untethered operation.
- Validating end-to-end system integration with PSLab mobile and desktop applications.

Team Vyadh - Mars Rover Team, VIT

Vellore, TN

Software Lead

April 2025 – Present

- Lead a multidisciplinary team of engineers, owning the rover's software and firmware architecture end-to-end; redesigned a fragmented control system into a unified, deterministic architecture with closed-loop velocity control and fault tolerance.
- Architected the fully autonomous outdoor navigation stack in **Python** using **ROS 2**; wrote a custom **C++ costmap plugin** for dynamic polygon-based keep-out zones.
- Designed a real-time ground control station telemetry pipeline over **MQTT**, streaming GPS/odometry, battery, environmental, and multi-spectral science-sensor data with fault-tolerant auto-reconnect handling.
- Built a 6-channel HD camera streaming pipeline in **GStreamer** with hardware-accelerated encoding for zero latency video, dynamic resolution scaling under bandwidth constraints, hot-plug/unplug camera support, and watchdog-monitored auto-reconnect, achieving **100%** streaming uptime across competition.

Antropi Robotics

Bengaluru, KA

Software Intern

May 2026 – June 2026

- Built a **6-DoF pose estimation** pipeline (**FoundationPose**, **SAM3**) in Python for a KUKA robotic arm enabling arbitrary object pick-and-place, and tuned an AMR navigation/docking stack (ROS 2) achieving **sub-1 cm** docking accuracy.
- Reverse-engineered proprietary CAD analysis features and built equivalent **automated manufacturing quoting** tooling in Python using OpenCascade.

PROJECTS

Hermes [GitHub] | Go, Redis Streams, Prometheus, Grafana, Docker

2026

- Built a pub-sub middleware in **Go** inspired by ROS 2/DDS internals; implements broker-managed topics, reliable **TCP** and best-effort **UDP** transports, **Redis Streams** durability with offset-based replay, JSON schema enforcement, node discovery, and **Prometheus**/Grafana observability.
- Used **OpenAI Codex** to scaffold the initial test suite, then hand wrote edge case tests for backpressure and dead letter queue handling.
- Sustained **256,000/256,000** reliable deliveries across **512 subscribers** with **48 ms p99 latency** and zero drops; reliable transport remained lossless through a **2,048-node** autonomous warehouse simulation while best-effort traffic degraded gracefully under load.

TECHNICAL SKILLS

Languages: C, C++, Python, Go, JavaScript

Systems & Distributed: TCP/UDP, Redis Streams, Prometheus, Grafana, MQTT, REST, WebSocket

Web & Backend: FastAPI, React, Docker

Robotics & Perception: ROS 2, Nav2, GStreamer, FoundationPose, SAM3, ArUco

Embedded: STM32, ESP32, RP2350, PIO, DMA, CAN, UART, USBFS

Tools: Git, Linux, CMake, GDB

AI Tools: Claude, OpenAI Codex, Github Copilot